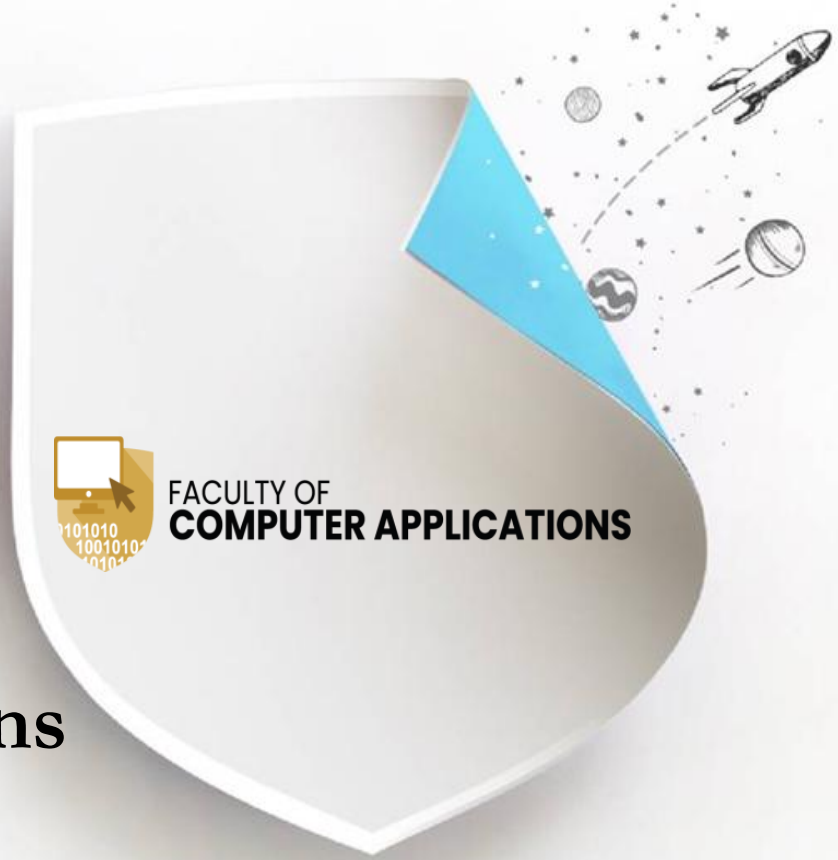
Marwadi University
Marwadi Chandarana Group

NAAC A+



FACULTY OF COMPUTER APPLICATIONS

Bachelor of Computer Applications

BCA Semester: 2

05BC3202 – Analysis and Design of Systems

PROGRAM	**BACHELOR OF COMPUTER APPLICATIONS**
SEMESTER	**2**
COURSE TITLE	**ANALYSIS AND DESIGN OF SYSTEMS**
COURSE CODE	**05BC3202**
COURSE CREDITS	**4**
COURSE DURATION	**20 HOURS**

UNIT 2 – INFORMATION GATHERING METHODS

TOPICS COVERED

INFORMATION GATHERING METHODS:

- Interviewing
- JAD
- Usage of questionnaire
- Sampling
- Investigation
- Observing decision makers' behavior and physical environment

TOPICS COVERED

AGILE MODELING AND PROTOTYPING:

- Prototyping
- Developing a prototype
- Rapid application development
- Agile modelling

INFORMATION GATHERING METHODS

1. Interviewing.
2. Joint application design (JAD).
3. Questionnaires.
4. Sampling.
5. Investigation.
6. Observing a decision maker's behavior.
7. Observing the physical environment.

1. INTERVIEWING

- An interview is a conversation between two or more people where the questions are from the interviewer to obtain information from the interviewee.
- Interview can be has two types: Interview of assessment and interviews for information
- The interview is the primary technique for gathering an information during system analysis phase.
- It is a skill which must be mastered by system analysts.
- Based on their skill it is determine the quality of information gathering and depth of it.
- Interviewing, Observation and research are the primary tools of the analyst.

Goals of interviee

- Gather Information about the company.
- Gather Information about the function.
- Gather Information of any process or activity
- To Uncover the problems
- Conduct needs determination
- Verification of previously gathered facts
- Gather opinions or viewpoints

Types of Interview

A. General

This is a one-on-one interview involving a reporter and an individual involved in a specific event or issue.



B. Talk Show

Many local television and cable stations **have interview shows where "people in news" are interviewed.** They are referred as **"soft" interviews** that usually focus on the **personality of the person or command**, rather on hard news issues.



C. Ambush

This type of **“on-the-run”, unanticipated interview** usually is related to some major issue or controversial event.



INTERVIEWING COMPONENTS

The interview process itself consists of a numbers of parts:

- Selection of the interviewee and scheduling time for the interview
- Preparation of interview questions, or script
- The interview itself
- Documentation of the facts and information gathered during the interview
- Review of the interview write up with the interviewee
- Correction of the write up, sign-off, and filing



INTERVIEWIN G GUIDELINES

INTERVIEWING GUIDELINES

1. First and foremost, establish the tone of the interview.
2. Let the interviewee know the reason for the interview and why he or she was selected to be interviewed.
3. Stress that the interviewee's knowledge and opinions are important and will aid in the analysis process.
4. Gain the interviewee's trust and cooperation early on and maintain it throughout.
5. Establish what will happen to the information gathered.
6. Determine any areas of confidentiality or restricted information.
7. Let the interviewee know that candor and honesty will be valued and that nothing will be published or passed on until it has been reviewed and verified by the interviewee.
8. Firmly establish that there are no negative consequences to being interviewed.

DOs of Interviewing

1. Do ask questions which start with who, what, where, when, why, and how, where possible.
2. Do ask both open and closed questions.
3. Do verify understanding through probing and confirming questions.
4. Do avoid confrontation.
5. Do act in a friendly but professional manner.
6. Do listen actively.
7. Do take notes, but do not be obtrusive about it.



DON'Ts of Interviewing

1. Do not assume anything.
2. Do not form pre-judgments.
3. Do not interrupt.
4. Do not go off on tangents.



2. JAD [JOINT APPLICATION DESIGN]

- Personal interviews are time-consuming and subject to error. So, an alternative approach to interviewing users one by one, called joint application design (JAD), was developed by IBM. The motivation for using JAD is to cut the time (and hence the cost) required by personal interviews and to improve the quality of the results of the information requirements assessment.

- Joint Application Development (JAD) is a collaborative methodology that brings together system developers and business stakeholders/end-users in structured, facilitated workshops to define requirements and design software applications

- It brings together business area people (end users) and IT professionals in highly focused workshops.

- JAD sessions are structured Workshop sessions

- Participants get together in a room to discuss the problem/project.

- Everyone hears what the rest of the group has to say.

PURPOSE OF JAD

- **Faster Development:** Reduced time for requirement gathering and sign-off.
- **Higher Quality:** Enhanced accuracy in requirements leads to fewer errors.
- **User Satisfaction:** Direct involvement ensures the final product meets user needs.
- **Cost Reduction:** Minimized rework and shorter development cycles.

WHO IS INVOLVED IN JAD

- Analyst (record stakeholder's decisions\, issues, Produce JAD plans etc)
- Stockholders (know business processes, challenges, know the system, requirements, critical issues, etc)
- Project Leaders
- Project Managers
- Project Sponsors

ADVANTAGES OF JAD

- Enhanced communication and relationship between business and end users and IT personnel.
- With the help of the JAD session, an organization can do the planning, receiving, receipt processing/tracking, monitoring and assigning, processing, recording, sending, and evaluating.
- For each topic, Organization asks questions about who, what, how, where, and why.
- Should also be asked and answered.
- Reduced system cost.
- Reduced development time.

ADVANTAGES OF JAD

- Improved system quality and productivity
- Design solutions
- Helps project teams get focused and stay focused
- Helps organization to get the right job done at the right time !!

DISADVANTAGES OF JAD

- JAD requires a two-to-four-day commitment, it is not possible to do any other activities in one day.
- It fails if preparation for the JAD session's report and documentation of specifications is incomplete OR inadequate (lacking quality).
- In these circumstances resulting designs could be less than satisfactory.
- Many variables need to come together correctly for JAD to be successful. Conversely, many things can go wrong.

3. QUESTIONNAIRES

- Questionnaires mean Surveys.
- A questionnaire consists of a set of questions to collect information from organizational people.
- It is an information-gathering technique that allows systems analysts to study attitudes, beliefs, and behaviour, of several key people in the organization affected by the current and proposed systems. who may be
- Attitudes are what people in the organization say they want (in a new system, for instance); beliefs are what people think is true; behaviour is what organizational members do.

3. QUESTIONNAIRES

- For the analyst, questions must be transparently clear, the flow of the questionnaire cogent, and it must be planned in detail.

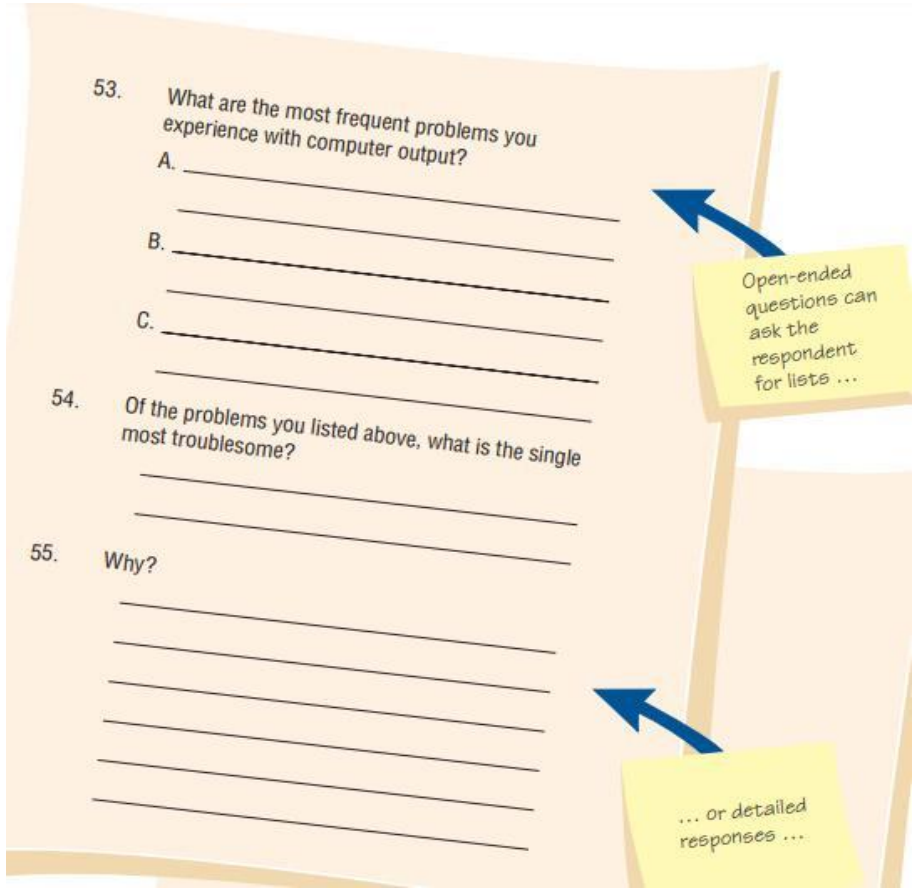
- **The basic question types used in the questionnaire are**

1. Open-ended
2. Closed-ended

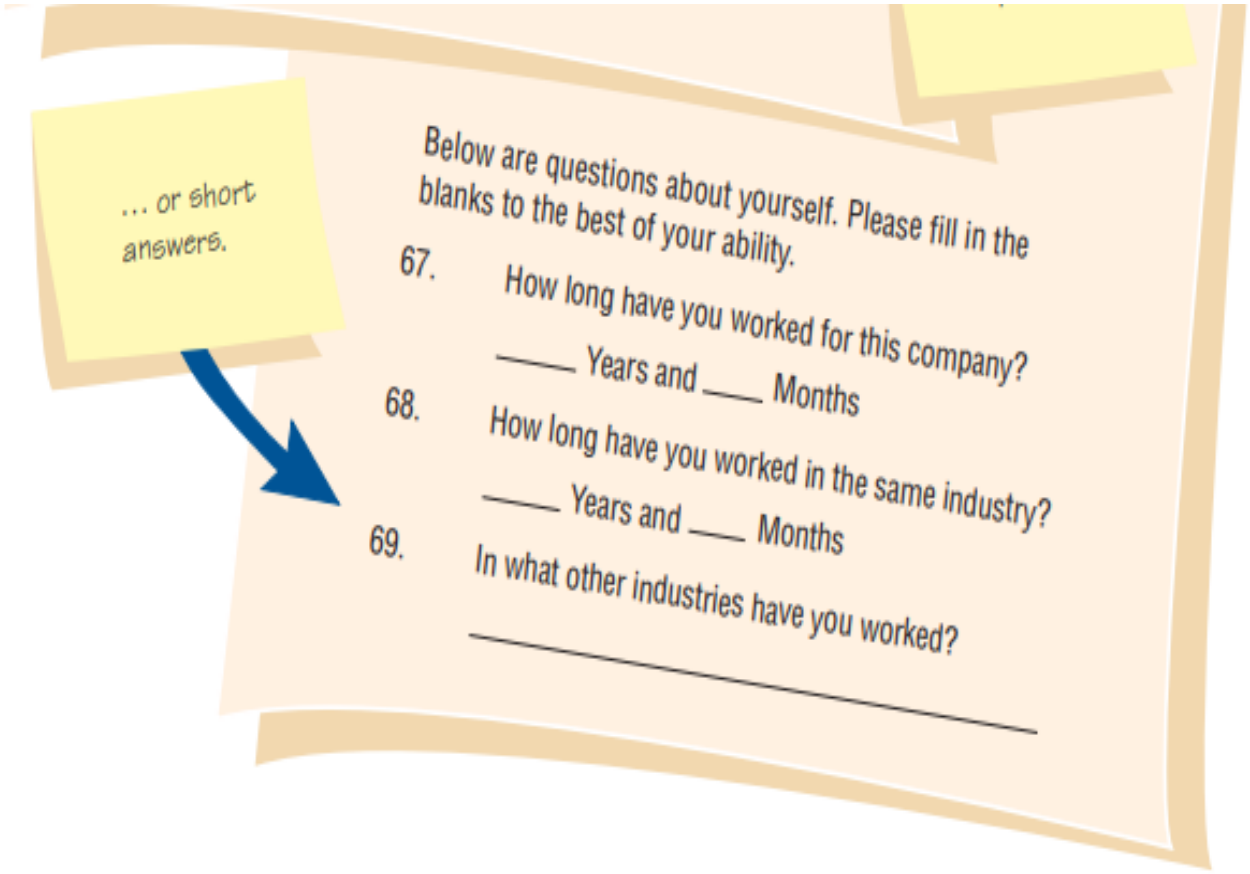
I. OPEN ENDED QUESTIONS

- "Open" actually describes the interviewee's options for responding. They are open. The response can be two words or two paragraphs.

- Some examples of open-ended questions are shown,



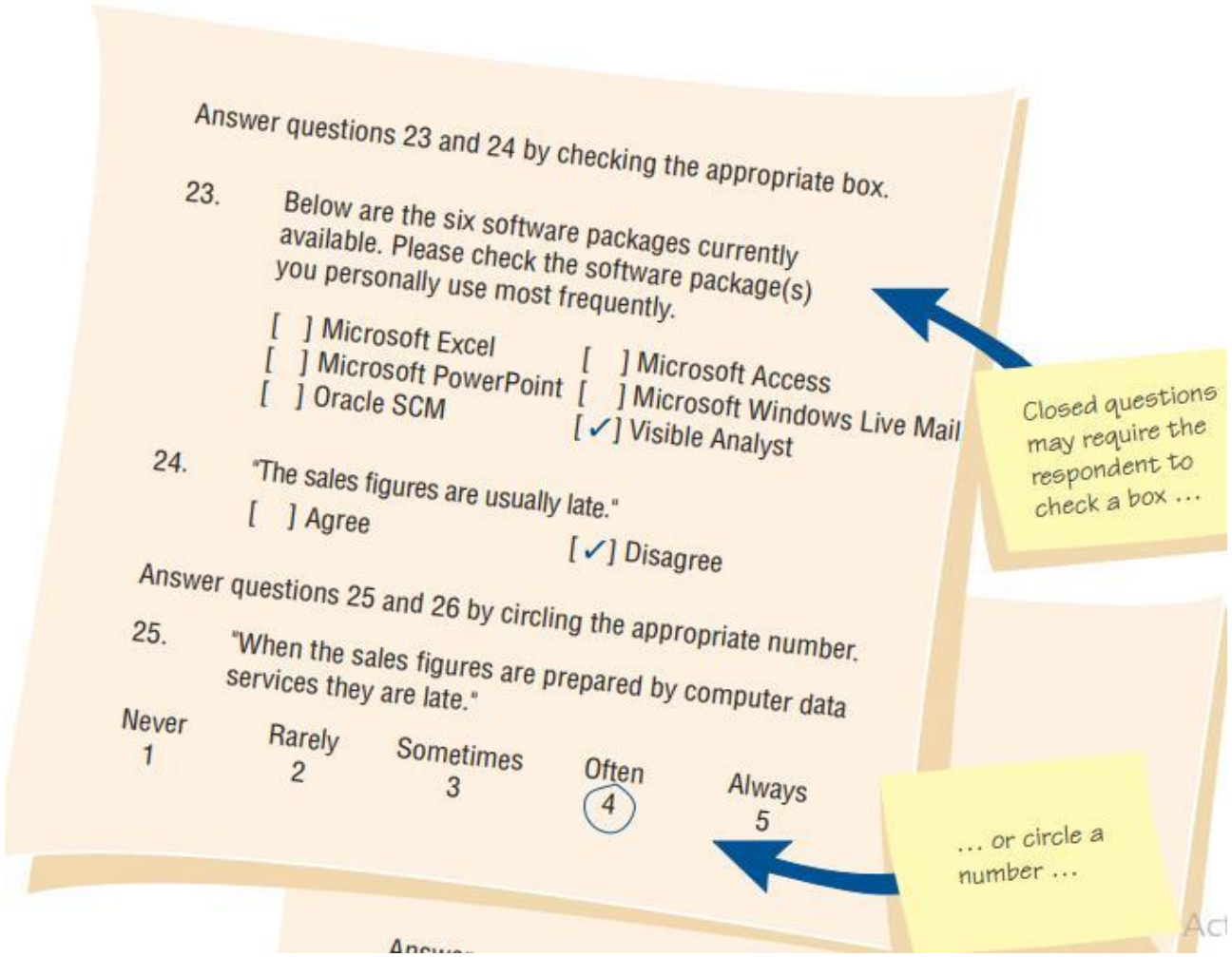
I. OPEN ENDED QUESTIONS



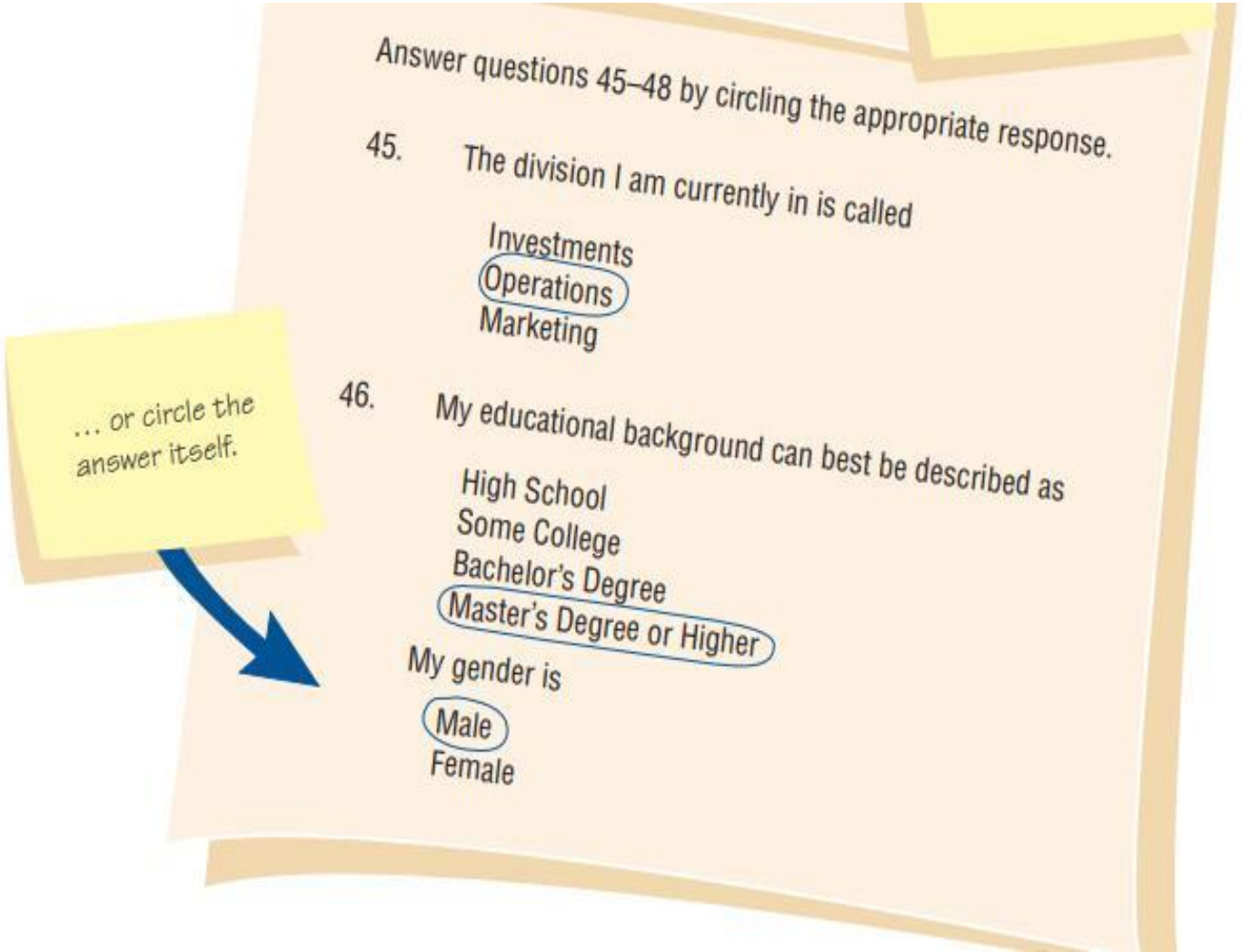
II. CLOSED ENDED QUESTIONS

- The alternative to open-ended questions is found in the other basic question type: closed questions.
- The possible responses are closed to the interviewee because he or she can only reply with a finite number such as “None,” “One,” or “Fifteen.”
- Some types of closed questions are MCQ, Rating, Bipolar(YES/NO, TRUE/FALSE, AGREE/DISAGREE ETC).

II. CLOSED ENDED QUESTIONS

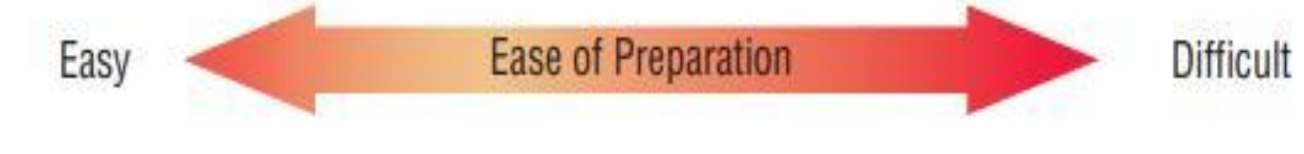


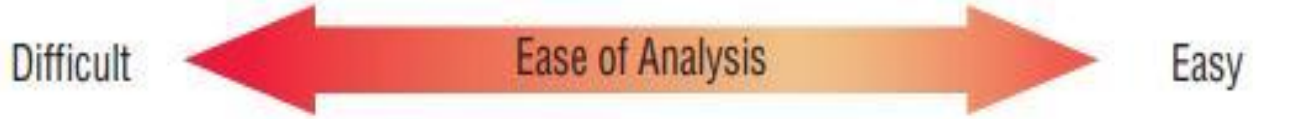
II. CLOSED ENDED QUESTIONS



OPEN ENDED VS CLOSED ENDED QUESTIONS



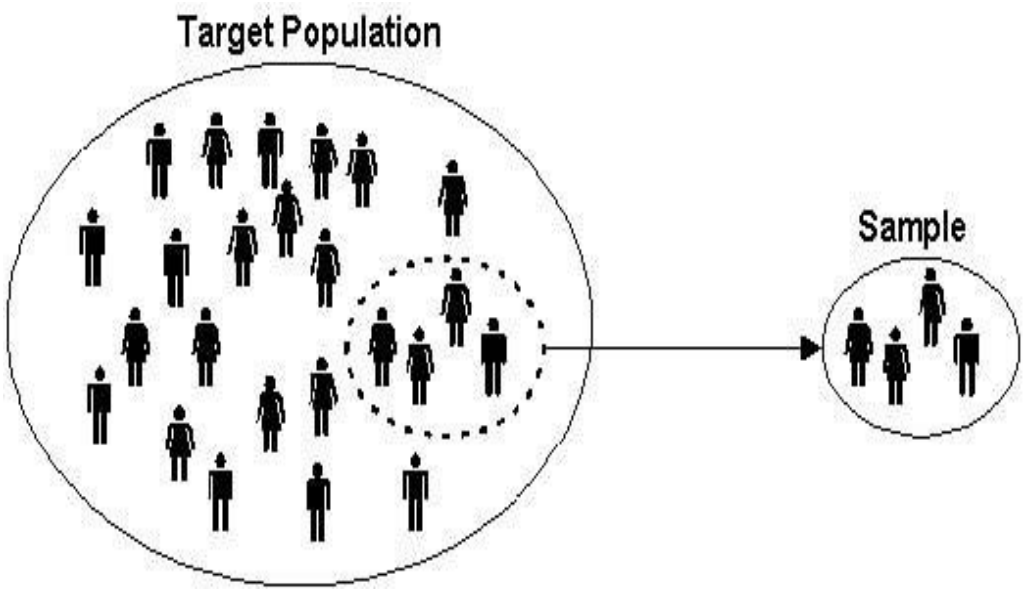




4. SAMPLING

- The process of systematically selecting representative elements of a population is called sampling.
- The purpose of sampling is to select and study documents such as invoices, sales reports, and memos, or perhaps to select and interview, give surveys to, or observe members of the organization.
- Reason For Sampling - Sampling can reduce cost, speed data gathering, make the study more effective, and possibly reduce the bias in the study.

4. SAMPLING



Ex 1



Ex 2 - Sample rice and lentils (based on samples one can buy it)

SAMPLING DESIGN STEPS

A systems analyst must follow four steps to designing a good sample:

1. Determine the data to be collected or described.
2. Determine the population to be sampled.
3. Choose the type of sample.
4. Decide on the sample size.

SAMPLING DESIGN STEPS

I. DETERMINING THE DATA TO BE COLLECTED OR DESCRIBED :

- The systems analyst needs a realistic plan about what will be done with the data once they are collected. If irrelevant data are gathered, then time and money are wasted in the collection, storage, and analysis of useless data.

- The duties and responsibilities of the systems analyst at this point are to identify the variables, attributes, and associated data items that need to be gathered in the sample. The type of data-gathering method (investigation, interviews, questionnaires, observation) is to be used. The kinds of information sought when using each of these methods are discussed in more detail.

SAMPLING DESIGN STEPS

II. DETERMINE THE POPULATION TO BE SAMPLED.

- Next, the systems analyst must determine what the population is. In the case of hard data, the systems analyst needs to decide, for example, if the last two months are sufficient, or if an entire year's worth of reports is needed for analysis.

- Similarly,the systems analyst has to determine whether the population should include only one level in the organization or all the levels.

SAMPLING DESIGN STEPS

III. CHOOSE THE TYPE OF SAMPLE.

- The systems analyst can use one of four main types of samples, They are convenience, purposive, simple, and complex.

- Convenience samples are unrestricted, nonprobability samples. A sample could be called a convenience sample if, for example, the systems analyst posts a notice asking for everyone interested in working with the new sales performance reports to come to a meeting at 1 P.M. on Tuesday the 12th. (Personal Meetings, Group meetings, selecting types with the help of Software tools, etc) Obviously, this sample is the easiest to arrange, but it is also the most unreliable.

SAMPLING DESIGN STEPS

IV. DECIDING ON THE SAMPLE SIZE

- If everyone in the population viewed the world the same way or if each of the documents in a population contained the same information as every other document, a sample size of one would be sufficient. Because that is not the case, it is necessary to set a sample size greater than one but less than the size of the population itself.

- It is important to remember that the absolute number is more important in sampling than the percentage of the population. We can obtain satisfactory results by sampling 20 people in 200 OR 20000 people in 2,000,000.

5. INVESTIGATION

An investigation is the act of discovery and analysis of data.

As the systems analyst works to understand users, their organization, and its information requirements, it will become important to examine different types of hard data that offer information unavailable through any other method of data gathering. Hard data reveal where the organization has been and where its members believe it is going. To piece together an accurate picture, the analyst needs to examine both quantitative and qualitative hard data

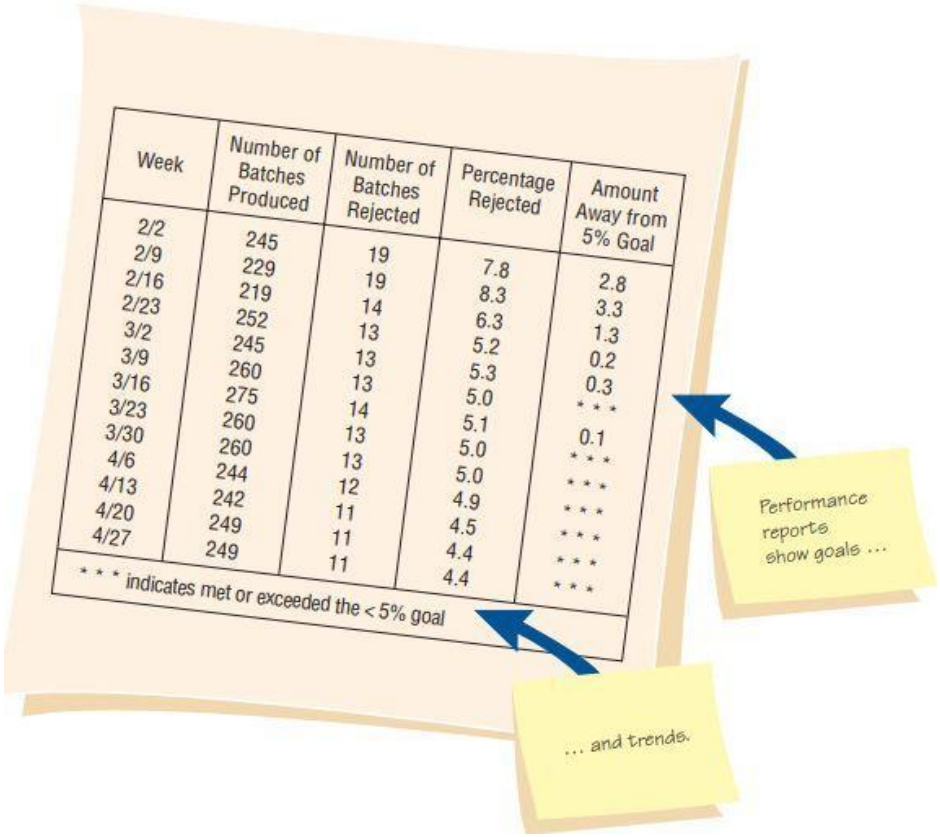
I. **Analysing Quantitative Documents (performance reports, records, and a variety of forms.)**

I. **Analysing Qualitative Documents (email messages, memos, slogans on work area, procedure manuals, policy handbooks etc)**

I. ANALYZING QUANTITATIVE DOCUMENTS

Many quantitative documents are available for interpretation in any business, and they include reports used for decision-making, performance reports, records, and a variety of forms.

A. REPORTS :

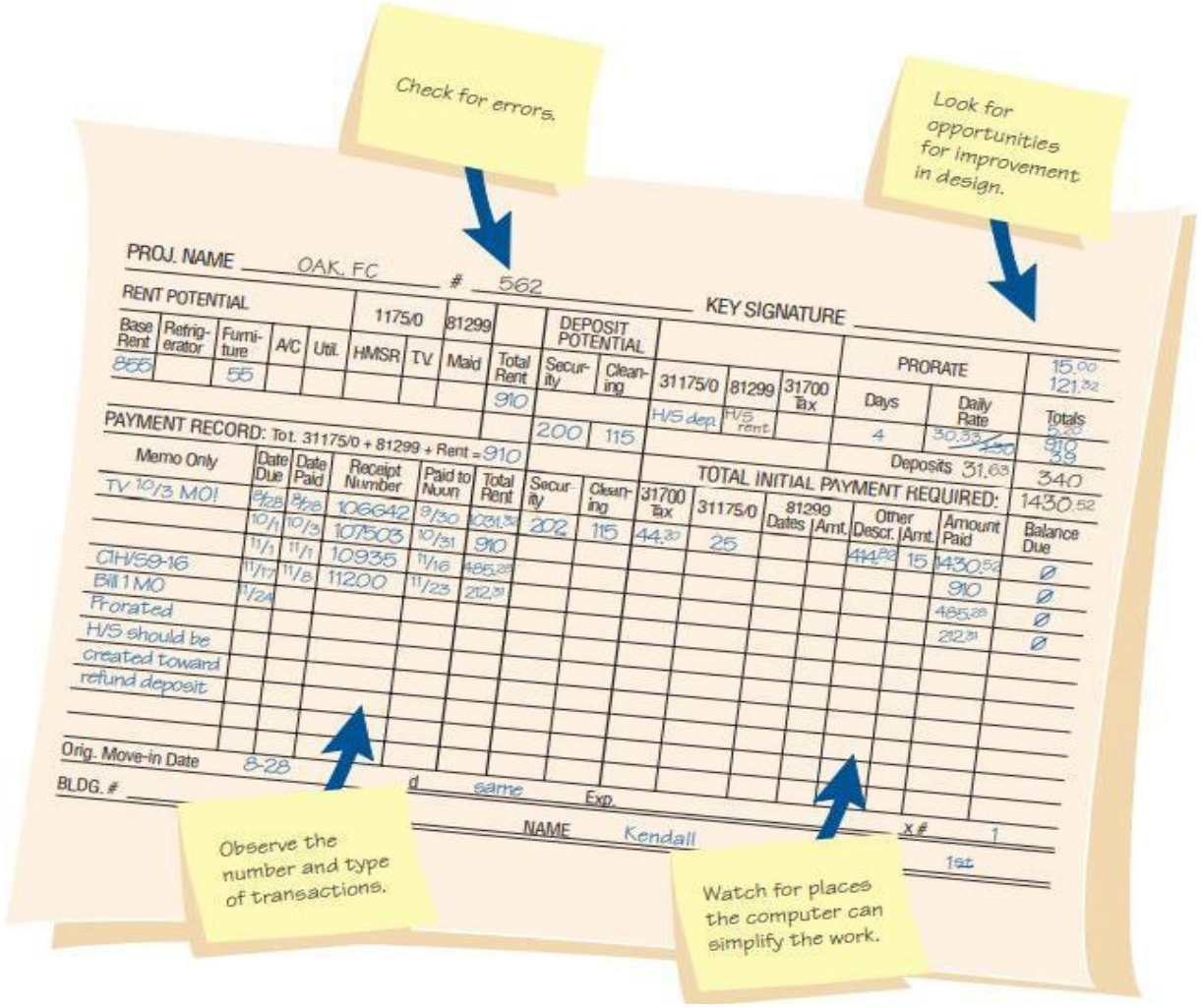


Week	Number of Batches Produced	Number of Batches Rejected	Percentage Rejected	Amount Away from 5% Goal
2/2	245	19	7.8	2.8
2/9	229	19	8.3	3.3
2/16	219	14	6.3	1.3
2/23	252	13	5.2	0.2
3/2	245	13	5.3	0.3
3/9	260	13	5.0	* * *
3/16	275	14	5.1	0.1
3/23	260	13	5.0	* * *
3/30	260	13	5.0	* * *
4/6	244	12	4.9	* * *
4/13	242	11	4.5	* * *
4/20	249	11	4.4	* * *
4/27	249	11	4.4	* * *

* * * indicates met or exceeded the < 5% goal

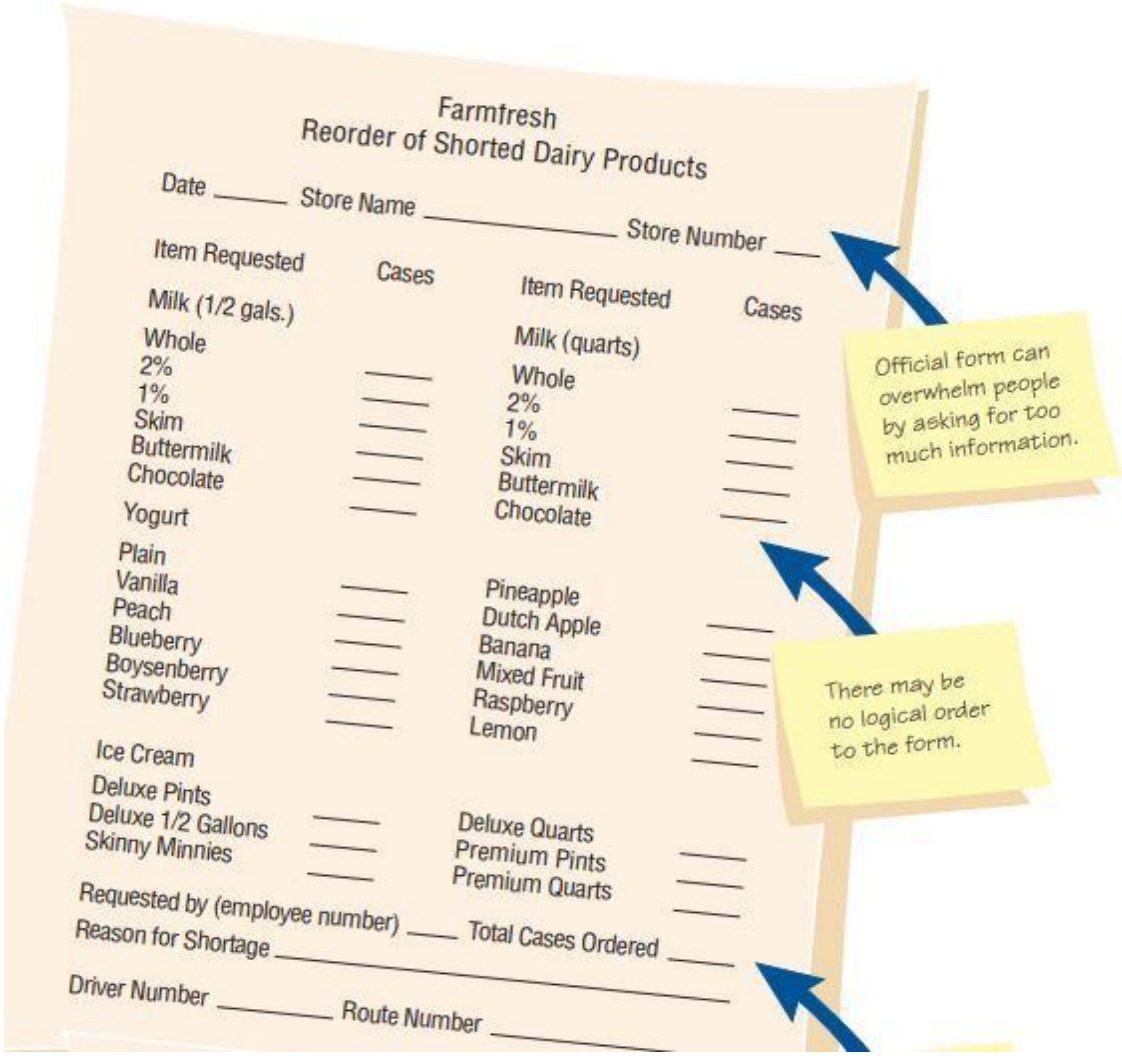
I. ANALYZING QUANTITATIVE DOCUMENTS

B. RECORDS:



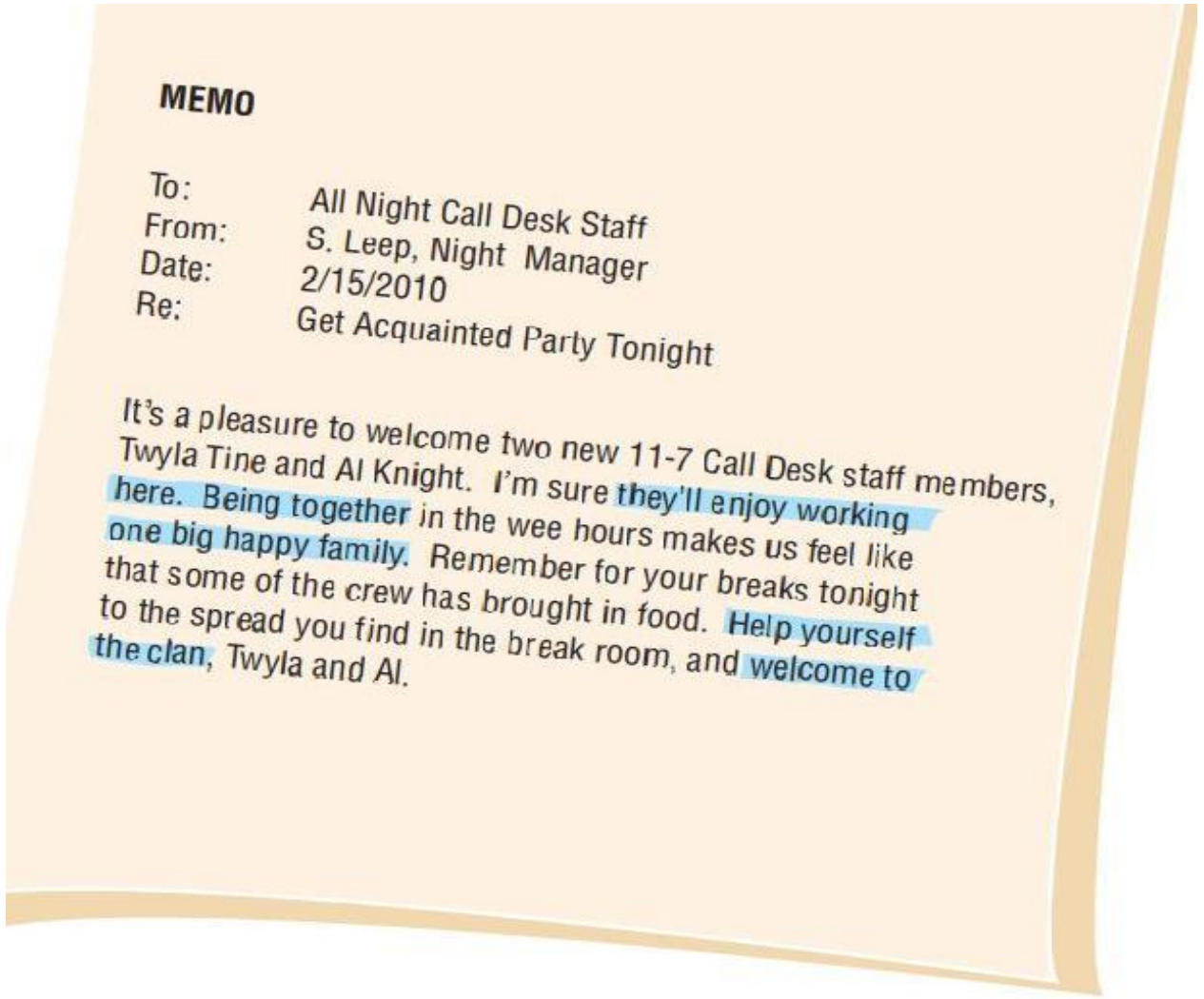
I. ANALYZING QUANTITATIVE DOCUMENTS

C. VARIETY OF FORMS:



II. ANALYZING QUALITATIVE DOCUMENTS

A. MEMOS :



MEMO

To: All Night Call Desk Staff
From: S. Leep, Night Manager
Date: 2/15/2010
Re: Get Acquainted Party Tonight

It's a pleasure to welcome two new 11-7 Call Desk staff members, Twyla Tine and Al Knight. I'm sure they'll enjoy working here. Being together in the wee hours makes us feel like one big happy family. Remember for your breaks tonight that some of the crew has brought in food. Help yourself to the spread you find in the break room, and welcome to the clan, Twyla and Al.

II. ANALYZING QUALITATIVE DOCUMENTS

B. SIGNS ON BULLETIN BOARDS :

Signs on bulletin boards and in work areas: Slogans posted such as "Quality Is Forever" or "Safety First" gives the analyst a sense of the official organizational culture.

C. CORPORATE WEBSITES:

Examine the contents for metaphors, humor, use of design features (such as color, graphics, animation, and hyperlinks), and the meaning and clarity of any messages provided. Think about the Web site from three dimensions: technical, aesthetic, and managerial.

II. ANALYZING QUALITATIVE DOCUMENTS

D. PROCEDURE MANUALS :

Manuals should be analysed following the five guidelines spelled out previously. Remember that manuals present the "ideal," the way machines and people are expected to behave.

E. POLICY HANDBOOKS :

Examining policies allows the systems analyst to gain an awareness of the values, attitudes, and beliefs guiding the corporation.

6. OBSERVING DECISION MAKER'S BEHAVIOR

- Observation, however, allows the analyst to see first-hand how managers gather, process, share, and use information and technology to get work done.
- Therefore, we suggest that systems analysts use a more humanistic approach to describe what managers do. This method is called the analyst's playscript.
- Playscript is an organized and systematic approach that demands the analyst be able to understand and articulate the action taken by each observed decision-maker.

6. OBSERVING DECISION MAKER'S BEHAVIOR

Playscript Analysis Company: Solid Steel Shelving Scenario: Quality Assurance
Analyst: L. Bracket Date: 1/3/2010

Decision Maker (Actor)	Information-Related Activity (Script)
Quality Assurance Manager	Asks shop floor supervisor for the day's production report
Shop Floor Supervisor	Prints out daily computerized production report
	Discusses recurring problems in production runs with quality assurance (QA) manager
Quality Assurance Manager	Reads production report
	Compares current report with other reports from the same week
	Inputs data from daily production run into QA model on computer
	Observes onscreen results of QA model
	Calls steel suppliers to discuss deviations from quality standards
Shop Floor Supervisor	Attends meeting on new quality specifications with quality assurance manager and vice president of production
Quality Assurance Manager	Drafts letter to inform suppliers on new quality specifications agreed on in meeting
	Sends draft to vice president via email
Vice President of Production	Reads drafted letter
	Returns corrections and comments via email
Quality Assurance Manager	Reads corrected letter on email
	Rewrites letter to reflect changes

7. OBSERVING THE PHYSICAL ENVIRONMENT

- Such observing means systematically examining the offices of decision-makers because offices constitute their primary workplace. There is one technique used in Structured Observation of the Environment (STROBE).

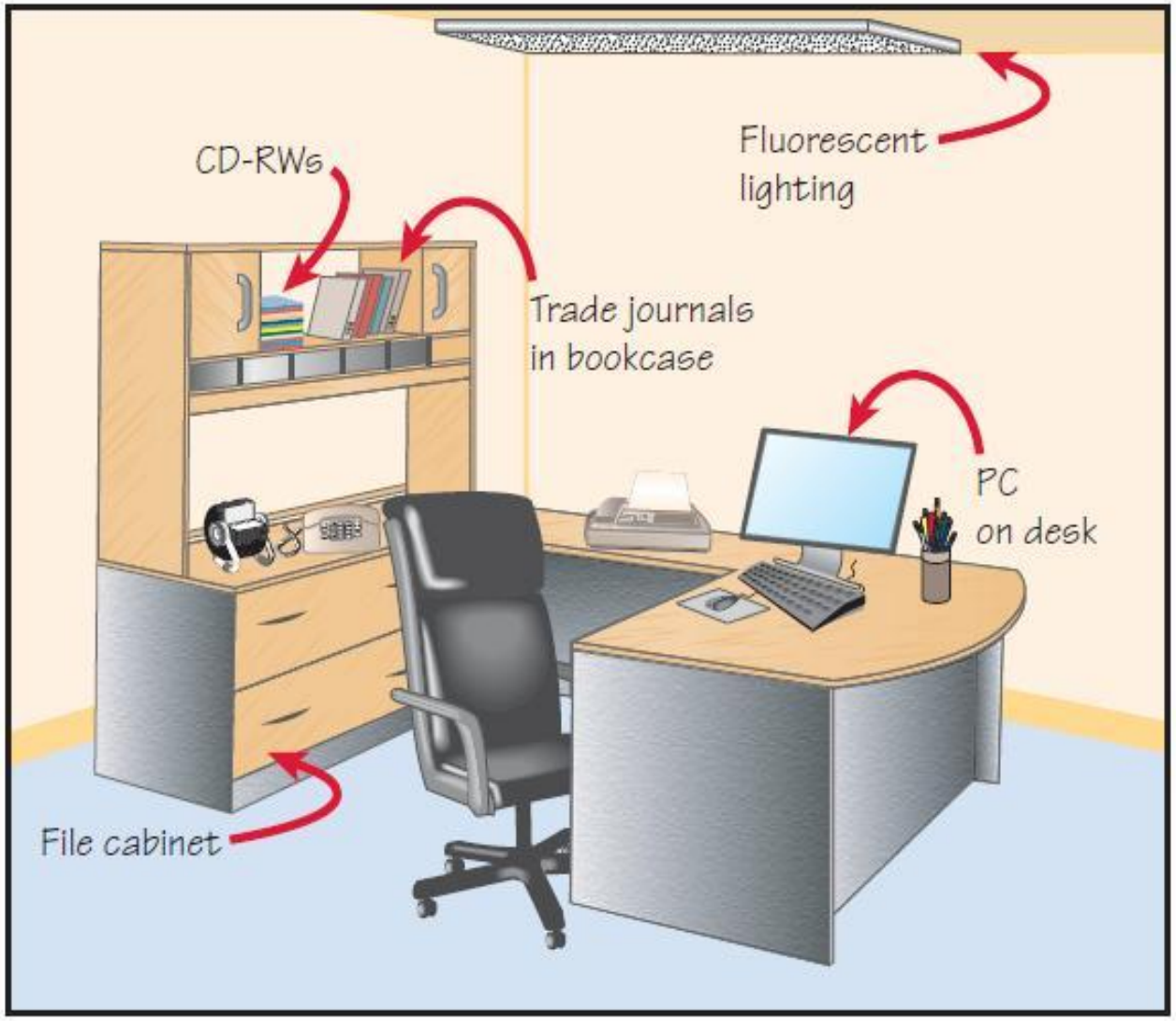
- The systems analyst should notice the arrangement of the office furniture and in particular the placement of the desk. shows an example of desk placement as well as many of the other elements of STROBE, such as props, stationary office equipment, lighting, color, and external sources of information.

- The method for Structured Observations of the Environment is referred to as STROBE. Successful application of STROBE requires an analyst to observe seven common elements in the offices.

7 ELEMENTS TO BE OBSERVED

Observable Element	Questions an Analyst Might Investigate
Office location	Who has the corner office? Are the key decision makers dispersed over separate floors?
Desk placement	Does the placement of the desk encourage communication? Does the placement demonstrate power?
Stationary equipment	Does the decision maker prefer to gather and store information personally? Is the storage area large or small?
Props	Is there evidence that the decision maker uses a PC, smartphone, or tablet computer in the office?
External information sources	Does the decision maker get much information from external sources such as trade journals or the Web?
Office lighting and color	Is the lighting set up to do detailed work or more appropriate for casual communication? Are the colors warm and inviting?
Clothing worn by decision makers	Does the decision maker show authority by wearing conservative suits? Are employees required to wear uniforms?

7. OBSERVING THE PHYSICAL ENVIRONMENT



AGILE MODELING AND PROTOTYPING

AGILE MODELING AND PROTOTYPING

- Prototyping
- Developing a prototype
- Rapid application development
- Agile modelling

PROTOTYPING

- It is a process in which developers create a model of the actual software. (Initial product is developed for demo purposes.)
- Prototyping of information systems is a worthwhile technique for quickly gathering specific information about users' information requirements.
- Purpose – To get the actual requirements more deeply from the user.
- Ex-beta versions.

TYPES OF PROTOTYPING

- Patched-up Prototype.
- Non-operational Prototype.
- First-of-a-series Prototype.
- Selected Features Prototype.

1. PATCHED-UP PROTOTYPE

- This is a working model with all the features but is inefficient.
- It has to do with constructing a system that works but is patched up or patched together.
- The retrieval and storage of information may be inefficient, however, because programs were workable rather than efficient.
- An example in information systems is a working model that has all the necessary features but is inefficient.

1. PATCHED-UP PROTOTYPE



2. NON-OPERATIONAL PROTOTYPE

- A non-operational prototype model is used when only a certain part of the program should be updated.
- Although it's not a fully operational program, the specific part of the program will work or could be tested as planned.
- An example of this approach is a full-scale model of an automobile that is used in wind tunnel tests. The size and shape of the auto are precise, but the car is not operational. In this case, only features of the automobile essential to wind tunnel testing are included.

2. NON-OPERATIONAL PROTOTYPE



3. FIRST-OF-A-SERIES PROTOTYPE

- A third conception of prototyping involves creating a first full-scale model of a system, often called a pilot.
- An operation model is Useful when many installations of the same information system are planned.
- An example is prototyping the first airplane of a series, then seeing if it flies before building a second.

3. FIRST-OF-A-SERIES PROTOTYPE



Success? YES.
Then make a series of airplanes



3. FIRST-OF-A-SERIES PROTOTYPE

- An example is a system to be installed in one location, tested and modified as necessary, and later implemented in other locations.
- Known as a beta version, this Prototype Model could be very efficient if properly launched.

4. SELECTED FEATURES PROTOTYPE

- A fourth conception of prototyping includes some, but not all, of the features that the final system will have.
- An analogy would be a new retail shopping mall that opens before the construction of all shops is complete.
- For example, users may view a system menu on a screen that lists six features: add a record, update a record, delete a record, search a record for a keyword, list a record, or scan a record.
- In the prototyped system, however, only three of the six may be available for use, so that the user may add a record (feature 1), delete a record (feature 3), and list a record (feature 5).

GUIDELINES FOR DEVELOPING PROTOTYPE

Guidelines for Developing a Prototype :

- Once the decision to prototype has been made, four main guidelines must be observed when integrating prototyping into the requirements determination phase of the SDLC :

1. **Work in manageable modules** (A manageable module allows users to interact with its features but can be built separately from other system modules.)

1. **Build the prototype rapidly** (Speed is essential for successful prototyping. Analysts can use prototyping to shorten this gap by finding information requirements)

GUIDELINES FOR DEVELOPING PROTOTYPE

3. **Modify the prototype in successive iterations** (Making a prototype modifiable means creating such modules that are not highly interdependent)

3. **Stress the user interface** (The user interface must be well developed enough to enable users to pick up the system quickly. Online, interactive systems using GUI interfaces are ideally suited

ADVANTAGES

The three major advantages of prototyping are the:-

1. Potential for changing the system early in its development,

1. The opportunity to stop development on a system that is not working,

1. The possibility of developing a system that more closely addresses users' needs and expectations.

- ❑ This way, the actual software could be released in advance. Everyone has to work on the same thing and at the same time, reducing man hours in creating a software

DISADVANTAGES

- Managing the prototyping process is difficult because of its rapid, iterative nature
- Since the prototype changes from time to time, it's a nightmare to create a document for this software. There are many things that are removed, changed, and added in a single update of the prototype, and documenting each of them has been proven difficult.

USER'S ROLE IN PROTOTYPING

Here are three main ways a user can be of help in prototyping:

1. Experimenting with the prototype.
1. Giving open reactions to the prototype.
1. Suggest additions to or deletions from the prototype.

AGILE

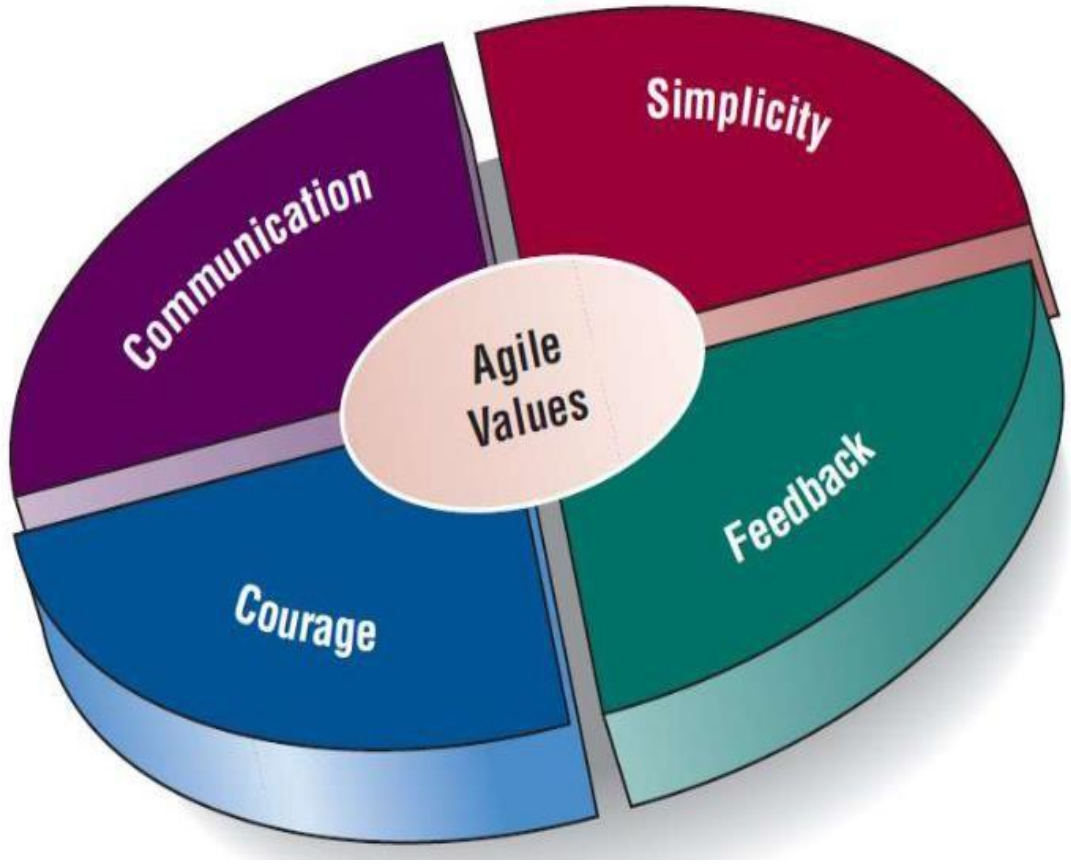
- Agile methodology is a way to manage the project by breaking it up into several phases
- Agile simply means continuous incremental improvement through small and frequent releases.
- The term Agile is most commonly associated with software development as a project management methodology.
- The agile approach is a software development approach based on values and principles.

The four values are communication, simplicity, feedback, and courage.



AGILE VALUES AND PRINCIPLES

- Communication
- Simplicity
- Feedback
- Courage





AGILE


Meet & Plan
Design
Code & Test
Release
Feedback
Agile
software
development


DELIVER
INCREMENTALLY
INSTEAD OF ALL
AT ONCE

BASIC PRINCIPLES

- ❑ Continue to deliver functioning software incrementally and frequently
- ❑ Encourage customers and analysts to work together daily.
- ❑ Promote face-to-face conversation
- ❑ Encourage continuous, regular, and sustainable development
- ❑ Adopt agility with attention to mindful design



BASIC PRINCIPLES

- ❑ Support self-organizing teams
- ❑ Provide rapid feedback
- ❑ Encourage quality
- ❑ Review and adjust behaviour occasionally
- ❑ Adopt simplicity



BASIC ACTIVITIES OF AGILE

Satisfy the customer through the delivery of working software, the below activities are required.

- ❑ Coding
- ❑ Testing
- ❑ Listening
- ❑ Designing



CODING

CODING :

- Coding is the one activity that it is not possible to do without
- The most valuable thing that we receive from code is “learning”
- Code can also be used to communicate ideas that would otherwise remain fuzzy or unshaped
- Source code is the basis for a living system. It is essential for the development



TESTING

TESTING :

- Testing is the second basic activity of development.
- This approach advocates writing tests to check the coding, functionality, and performance.
- Agile modeling relies on automated tests, and large libraries of tests exist for most programming languages
- These tests need to be updated as necessary during the progress of the project.



LISTENING

LISTENING:

- The third basic activity of development is listening.
- In the agile approach, listening is done in the extreme.
- Developers use active listening to hear their programming partners and customers.
- Developers have to assume that they know nothing about the business so they must listen carefully to business people.



DESIGNING

DESIGNING:

- The fourth basic activity in development is designing, which is a way of creating a structure.
- Organize all the logic in the system. Designing is evolutionary, and so are systems that are designed.



RESOURCE CONTROL VARIABLES

- **TIME:** You need to allow enough time to complete your project. You need time to listen to the customers, time to design, time to code, and time to test.
- **COST:** Cost is the second variable we can consider adjusting
- **QUALITY:** The third resource control variable is quality. It allows the analyst to put less effort into maintaining quality.
- **SCOPE:** Finally, there is scope. In the agile approach, the scope is determined by listening to customers and getting them to write down their stories



RESOURCE CONTROL VARIABLES

Need or Opportunity:	Apply shortcut methods for faster checkout.					
Story:	If the identity of the customer is known and the delivery address matches, speed up the transaction by accepting the credit card on file and the rest of the customer's preferences such as shipping method.					
		Well Below	**Below Average**	**Average**	**Above Average**	**Well Above**
Activities:	**Coding**			✓		
	Testing			✓		
	Listening			✓		
	Designing				✓	
Resources:	**Time**				✓	
	Cost		✓			
	Quality					
	Scope			✓	✓	



FOUR CORE AGLIE PRACTICES

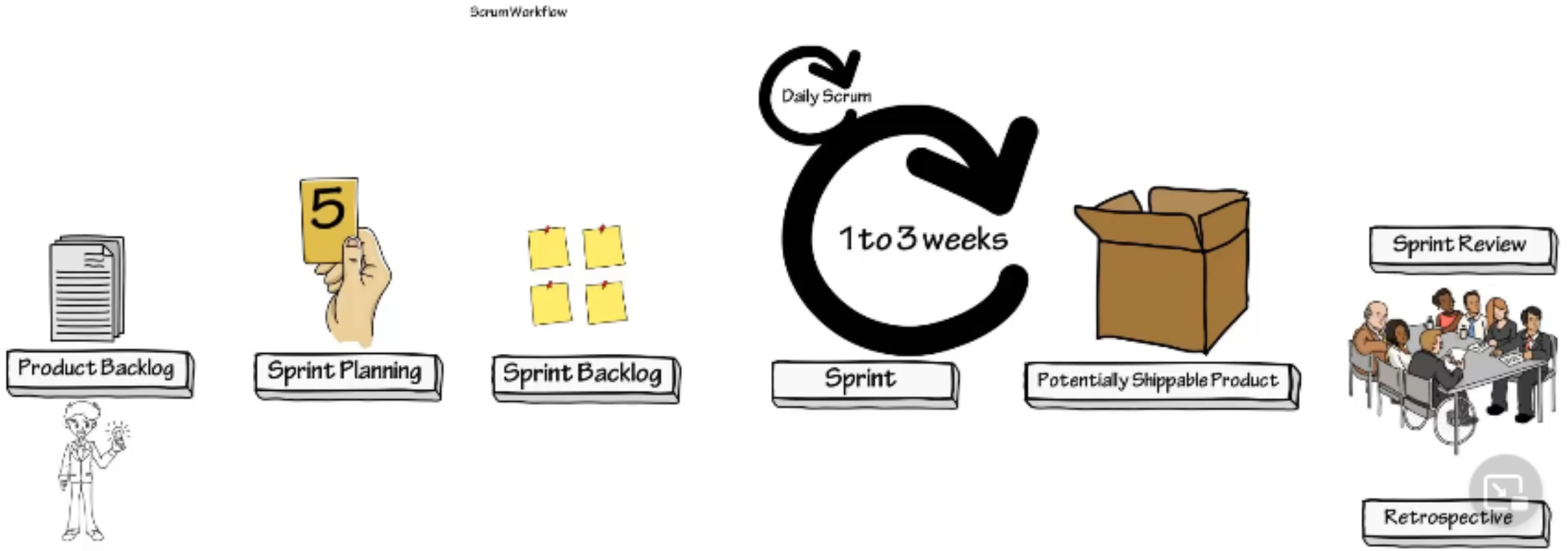
- ❑ Short releases
- ❑ 40-hour work week
- ❑ Onsite customer
- ❑ Pair programming



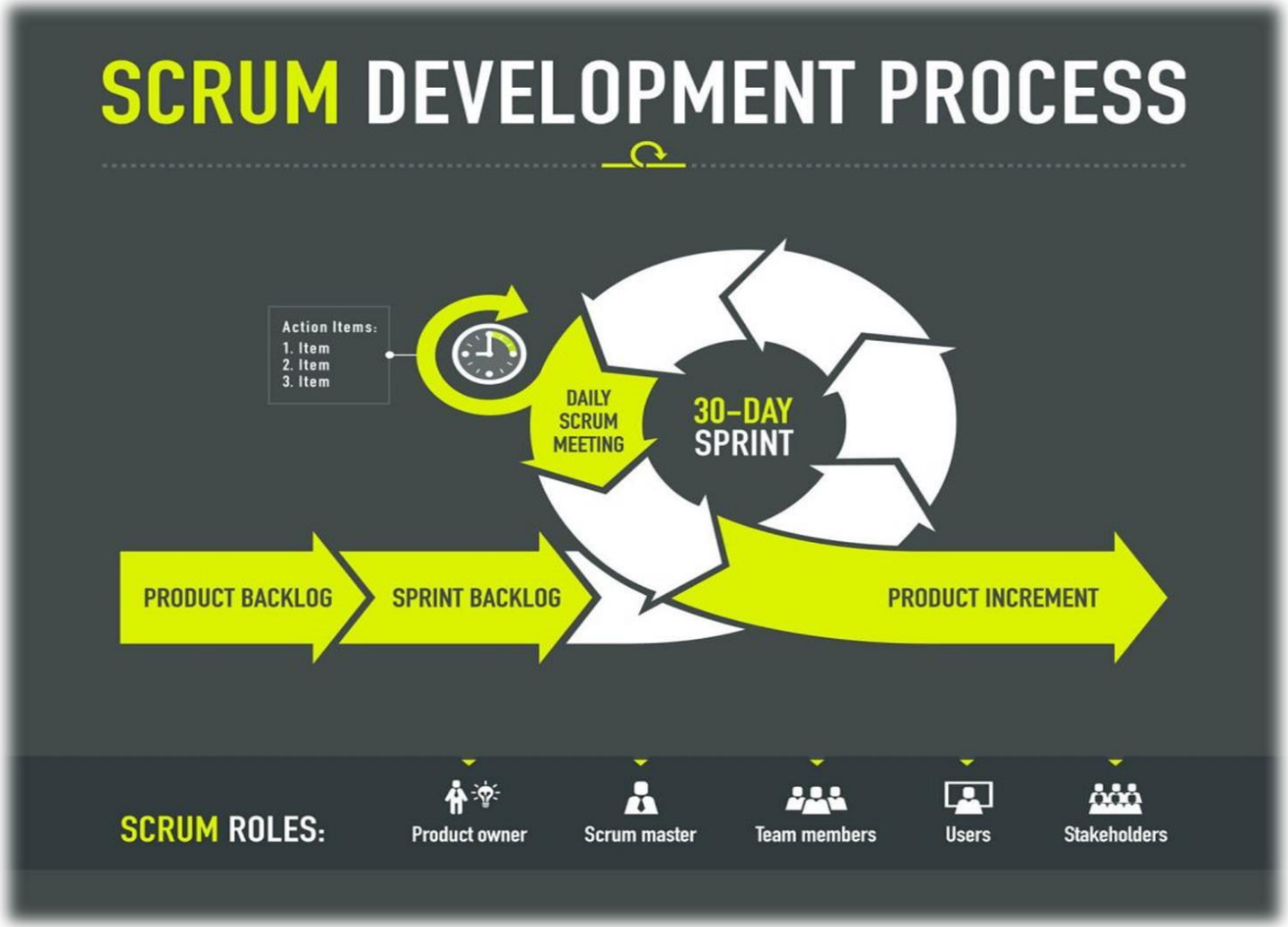
SCRUM

- Scrum is a framework of rules, roles, events used to implement Agile projects.
- It is an iterative approach, consisting of sprints that typically only last one to four weeks. This approach ensures that your team delivers a version of the product regularly. Scrum was designed using a software model that follows a set of roles, responsibilities, and meetings. It can be used for any complex project but works best when the result is a concrete product rather than a service.

SCRUM


Scrum Workflow
Product Backlog
5
Sprint Planning
Sprint Backlog
Daily Scrum
1 to 3 weeks
Sprint
Potentially Shippable Product
Sprint Review
Retrospective
Repeat this workflow for each sprint

SCRUM


SCRUM DEVELOPMENT PROCESS
Action Items:
1. Item
2. Item
3. Item
DAILY SCRUM MEETING
30-DAY SPRINT
PRODUCT BACKLOG
SPRINT BACKLOG
PRODUCT INCREMENT
SCRUM ROLES:
Product owner
Scrum master
Team members
Users
Stakeholders

SDLC VS AGILE

S D L C	A G I L E
SDLC is a process of design and development of a product or service.	Agile is a methodology following an iterative approach used for project management purposes.
Software Development Life Cycle was foundin 1960.	It came out in 2001 and it was made by a joint collaboration of 17 software developers.
SDLC has a systematic approach.	Agile has a faster approach comparatively.
SDLC doesn't allow changes after initialstage.	Agile allows dynamic changes inrequirements.
Teams find bugs only after the completedevelopment cycle.	Constant testing keeps bugs in check.
Changes not allowed after planning phase.	Changes are allowed any time.
SDLC projects deliver as a complete package.	Delivers small pieces of larger projects with each Spring.
Structure is Rigid.	Structure is flexible.

THANK YOU

